MALMÖ UNIVERSITY

Speculative Futures of AI in Art

Collaborative Design Fiction with Artists

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Abstract

As generative AI threatens creatives worldwide, this thesis applies Speculative Design through a Participatory Design process to speculate about the futures of AI in art by critically involving creatives. Conducting field research, interviews, and two co-design workshops, hosted at the Malmö City Library, the project resulted in an exhibition of four speculative scenarios in the form of short stories and AI-generated visualisations, sparking critical discourse and reflection about generative AI tools in art. The main insights from said discourse were that AI tools need to be investigated and critiqued through use by creatives, that designers working on AI tools have a responsibility to design for transparency, and that Speculative Design is the appropriate methodology to address AI in art, especially grounded in a Participatory Design process.

Keywords: Interaction Design, Human-Computer Interaction (HCI), Participatory Design, Speculative Design, Artificial Intelligence (AI), Design Fiction

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1 Introduction

1.1 Context

The most recent wave of generative AI tools has gathered a great amount of attention throughout society. By some, AI is seen as humanity's saviour and by others, it is seen as humanity's doom. While we are far away from either extreme, generative AI tools are widespread with great numbers of users. OpenAI's ChatGPT is the fastest-growing consumer application in history, reaching 100 million users in just two months (Hu, 2023). ChatGPT generates text in response to prompts, be it articles, essays, jokes or even poetry. With other tools, other modalities followed, including image, video, and sound generation now available to the public.

Those thought to be safe from automation, creatives, seem to have been impacted the most by generative AI. Writers, visual artists, videographers, and musicians find themselves in a world in which their work is compared to material generated in seconds by AI-powered tools. This has led to a devaluation of creative practice, creating uncertainty of employment for many artists. Additionally, AI models are competing with artists, while they are often trained on copyrighted data, using artist's work without permission.

Furthermore, the term "AI-Art" has sparked discussion among traditional artists and AI enthusiasts. Refusing to accept AI-generated material as art, some artists point out that there is no craft, no artistic process, and no human emotion in such outputs, while enthusiasts answer that refining prompts to visualise ideas instantly using generative AI empowers their creativity as no other tool has. It follows that there is ongoing debate about what constitutes art in the age of generative AI.

Since the topic of AI in art touches on complex themes such as individual perception of creativity and ownership, and societal structures of business and technological innovation, it can be considered a wicked problem, which is a symptom of multiple complex or even conflicting issues (Marshall, 2008). Addressing such a wicked problem through a traditional design process will not provide any meaningful results. Traditional user-centred design lacks in its ability to question and change our values (Dunne & Raby, 2013), which is needed to reach beyond the superficial when working with AI in art.

Speculative Design (SD) is the appropriate design perspective to apply when dealing with wicked problems. Since AI in art is a new and ill-defined phenomenon, there is a need for reflection, critique, provocation, and inspiration, which can be facilitated through SD. Critical discussion and discourse about AI in art is needed. Thus, this project applies a SD perspective.

For meaningful speculation, it is crucial that the people who are affected by AI are given a say in the design process. As established, earlier artists are particularly impacted. Enabling artists' participation, collaborative speculation promises a critical perspective on AI in art. Adding a Participatory Design (PD) perspective, artists were involved to have an influence on the design process.

Combining SD and PD, the design process started with field research and interviewing artists. An exploratory workshop was held to facilitate discussions among artists about AI in art and society. Furthermore, the main event of the project was a collaborative speculation workshop, practising design fiction with artists. The results were reworked and exhibited at the city library at a final exhibition.

1.2 Aim & Research Question

Applying SD, the aim of this project is to spark discussion and reflection about generative AI in art, while enabling meaningful participation of artists throughout the design process.

In this project, I work with one research question and two sub-questions. The main research question focuses on the designerly aspect of the project, which is to design speculative artefacts. It is stated as follows:

How might we speculate through design about the relationship between AI, art, and creatives in society in the present and future?

The two sub-questions address empirical and methodological themes. The empirical sub-question grounds my qualitative research and the methodological sub-question focuses on the design process of the project combining two types of design. They are stated as follows:

1. *How do creatives see AI in relation to art and society and what are the implications?*

2. *How can Speculative Design be facilitated in a Participatory Design process to spark critical debate about AI in art among creatives?*

1.3 Contribution

This project resulted in a set of scenarios and speculative artefacts that are presented in an exhibition at the Malmö City Library, which can be seen as a research outcome, in line with the notion of performative research, as defined by Haseman (2006). The scenarios of speculative futures and fictional objects that exist within them are presented in the form of short stories and visualisations. The short stories were handwritten, while the visualisations were AI-generated. These scenarios and fictional artefacts were developed by participants in a speculative co-design workshop. To spark discussion and reflection about the implications of AI in art and society, the scenarios outline speculative futures, through the short stories, while the AI-generated

visualisations provoke reactions and a concrete discussion about the use of generative AI in art and its implications for artists.

Furthermore, my qualitative research led to insights into how creatives and artists see AI in art and society and its implications. This can be seen as an additional contribution by itself, while also informing the SD process.

Additionally, I contribute to Interaction Design methodology by conducting a SD project with a PD process. I reflect and discuss my methods and the design process.

1.4 Collaboration with a Public Library

The project was a collaboration with the Malmö City Library. The library acted as a host for the two workshops and for a final exhibition of the resulting scenarios and artefacts. The first workshop, which focused on a critical discussion between artists about generative AI, is set in the Media Lab, a creative space within the city library. The second workshop, which employs SD, was part of a series of lectures and workshops about AI, called “Nyfiken pa AI”, and is set in an open space at the Library. The exhibition was similarly displayed openly at the library, making it available to the public.

The library has a stake in the project, because of its mission to inform citizens, and because it is a natural meeting place for artists and creatives. The workshops were available to the public and required no registration or attendance fee. They were located in a way that library visitors could spontaneously join the workshops.

1.5 Structure of the Thesis

The thesis is structured as follows. In the background section, relevant themes of AI and artists are expanded upon in addition to PD and SD. These design approaches are picked up again in a general methods section. This is followed by three sections for activities of the project. These sections contain their own methods, results and discussion to give the reader an easy understanding of how the activities informed each other. Next, the design proposal is presented, followed by the discussion section engaging in a variety of relevant themes, addressing the research question and sub-questions. Finally, the conclusion of the thesis is presented.

2 Background

2.1 Artificial Intelligence

Currently, the term AI can be heard loud and often, however, there is no one clear definition. Rather, the definition of AI depends on the area of research in which the term is used (Wang, 2019). In the project, I mainly address generative AI, which refers to AI models capable of generating new content based on existing data. Currently, a variety of modalities can be generated by different AI models, such as text, image, video, sound, speech or code.

In the following paragraph, I will give a non-technical explanation of how generative AI works. However, the explanation is largely simplified to not overcomplicate this section. Generative AI models use neural networks, like transformers, that use layers that can process and understand the context within data sequences (Mitchell, 2019). The models are trained on large datasets. Learning from patterns, they predict the next token in a sequence. A token can be a piece of a word, in text generation, a group of pixels, in image generation, or a fraction of sound, in sound generation (Mitchell, 2019). Given an initial prompt, often text, as input, the model can generate an output that statistically fits the input. On the generated output, models are fine-tuned to increase quality (Mitchell, 2019).

The current period of accelerated progress in generative AI, often referred to as AI spring, has gathered a lot of public attention through the release of tools like ChatGPT, for text generation, DALL-E, for image generation, or Sora, for video generation. Said AI tools are all developed by OpenAI, partly owned by Microsoft. Companies like Google, Meta, Anthropic and Mistral have released their own competing tools.

AI companies have repeatedly been involved in controversies, refusing to publish the exact origins of their training data (Stark & Hoffmann, 2019). These companies train their privately held AI models on publicly available data from all over the internet, including copyrighted data, and claim that it is fair use (Crawford, 2021). While the extraction of public knowledge value to increase private profits is common for big tech companies and capitalism as a whole, this latest manifestation takes an unprecedented scale and disregard for existing copyright laws (Crawford, 2021).

2.2 Artists

In this project, when referring to *artists* or *creatives*, I am addressing any person who engages in any creative activity, for example, visual graphics artists, performance artists, musicians, filmmakers, actors, creative professionals, and fine artists.

Most famously, American actors went on strike to demand, among other things, the protection against the unregulated use of AI and digital replicas without consent or compensation (SAG-AFTRA, 2023). This was met by vocal support from creatives throughout the world.

The disregard for the intellectual property of AI companies, as outlined above, has led to the emergence of anti-AI tools, which led artists to protect their artwork from being exploited by AI tools, mostly image generators. “Glaze” is a tool that changes artists’ images in a way that is not perceivable to humans but misleads AI image generation models that try to reproduce the artists’ style (Shan, Cryan, et al., 2023). Furthermore, the tool “Nightshade”, has the ability to poison an AI image generation model, through altered images, by corrupting its ability to respond to specific prompts (Shan, Ding, et al., 2023). This is presented as a protection against web scrapers who disregard the artists’ intellectual property.

2.3 Participatory Design

As emphasized above, the methodology of the project is grounded in PD principles. PD, also referred to as Co-Design, is characterized by the critical involvement of relevant stakeholders throughout the design process, to ensure that the resulting solutions meet the user’s needs (Simonsen & Robertson, 2013). Values of democracy and emancipation are fundamental to PD (Simonsen & Robertson, 2013). Besides the users, the relevant stakeholders, which depend on the nature of the given project, can include for example company managers, municipality officers or software developers. Common methods of PD include collaborative activities such as workshops or prototyping sessions (Simonsen & Robertson, 2013). PD is a firmly established area of research and is practised globally within the Human-Computer Interaction (HCI) community (Simonsen & Robertson, 2013).

PD originated in Scandinavia during the workers’ rights movement in the 1970s. More specifically, it emerged as a reaction to the introduction of computer-based systems to the workplace and the corresponding changes (Simonsen & Robertson, 2013). Early projects like DEMOS and UTOPIA served as examples of how the collaboration of researchers and trade unions could lead to successful design (Ehn, 1993). A notable figure in the field, Pelle Ehn was involved in said projects and further developed the field of PD. Ehn (1993) names two central aspects of PD, its political approach of democracy, power, and control in the workplace, and PD as a way of practising successful design through the participation of skilled users in the design process.

There is an urgent need for PD in the face of dramatic changes in connection to technology and corporate influence, as Bødker and Kyng (2018) state. However, the current practice of PD finds itself occupied in small-scale projects with limited time frames (Bødker & Kyng, 2018). PD ought to face *big issues*, as Bødker & Kyng (2018) state, by taking on larger-scale projects, making lasting alliances, and embracing high technological ambitions

through working prototypes. Addressing *big issues* means for people to "take control and partake in the shaping and delivery of technological solutions" that affect their lives, as Bødker and Kyng (2018, p. 4:1) state. Furthermore, there is an increasing awareness of the politics that are involved in the preparation, negotiation, and planning of PD, which Pedersen (2016) calls *design before design*.

Moreover, one of the contemporary *big issues* seems to be AI. Accordingly, the term Participatory AI has gained momentum, along with promises of safe, legitimate, and ethical AI systems that rely on participation (Birhane et al., 2022). However, these claims seem to be based on a misunderstanding of participation and PD principles, which could even lead to the exploitation of participants, as Birhane et al. (2022) point out. Yet, there is an ongoing debate on how genuine PD can shape the design and development of AI systems for the better (Zytko et al., 2022).

2.4 Speculative Design

SD is a way of designing that explores the potential futures of society through provocative and imaginative design artefacts, often through speculative scenarios of alternative futures (Dunne & Raby, 2013). Rather than designing solutions, as done in traditional design, SD is concerned with asking questions. It aims to spark discussion and reflection about the broader implications of new technologies in society (Dunne & Raby, 2013).

First defined by Dunne and Raby (2001) in their work "Design Noir: The Secret Life of Electronic Objects", SD has since established itself in the field of design. Other notable works include "The Presence Project" by Gaver and Dunne (2001) and most famously "Speculative Everything: Design, Fiction, and Social Dreaming" by Dunne and Raby (2013).

Due to the often conceptual and abstract nature of SD, it has been criticised as inaccessible and catered towards a niche audience of designers and academics, making it elitist (Mitrovic et al., 2021). However, there has been an emergence of participatory approaches to SD addressing the criticism, as Mitrovic et al. (2021, p. 81) note, "the participatory approach opens up possibilities for people to think about, imagine but also to act in creating their preferable future".

While this project applies such a participatory approach to SD, Farias et al. (2022) have outlined a framework for assessing the level of engagement of participants from non-participation to leadership.

3 Method

3.1 Methodological Approach

The project's mode of inquiry combines SD and PD. As mentioned previously, SD is applied to the project because of its ability to critically address wicked problems, which are characterized as being symptoms to multiple complex interconnected issues (Marshall, 2008). The technology of generative AI is rapidly developing and novel. Additionally, it touches on individual values regarding creativity and ownership, and societal structures of business and technology innovation. Thus, the expanding use of generative AI and its impact on artists is a wicked problem, best addressed through SD.

As outlined above, traditional PD is applied, because involving users through the entire design process should ensure that a design meets the users' needs. However, this project is not aiming to design something that meets the users' needs in the form of a solution, but to design something that can question and change the users' values, beliefs and behaviours. To achieve this through SD, the direct involvement of users, as seen in PD, is imperative, because only a design grounded in the experiences of artists and creatives can result in authentic speculation about the relationship between AI, art, and creatives in society, as articulated in my research question.

Due to this methodology, the design process deviates from the classic double diamond model, but an initial research phase followed by a design phase is still present. However, the research phase extends further into the project and the design phase is largely dependent on the speculative workshop activity.

3.2 Grounded Theory

I apply grounded theory throughout the project to make sense of qualitative data gathered during fieldwork, interviews, as well as exploratory and generative workshops. Grounded theory, as established by Glaser and Strauss (1967), is a research approach that starts with qualitative data gathered by researchers, which is used to develop a new theory. While originating as a social science tradition, the method of grounded theory can also be applied in design research when working with qualitative data, as Muratovski (2021) states.

Applying grounded theory, starting with a research question, the first step is to collect initial qualitative data (Charmaz, 2014). Coding the data is a central aspect of the method. Charmaz (2014, p. 46) states that, through coding, researchers "define what is happening in the data and begin to grapple with what it means". Initial Coding is the first step of labelling data openly and can range from word-by-word coding to a more general approach of coding loosely within the data (Charmaz, 2014). The second phase is focused coding,

which is used to “synthesize and explain larger segments” of data by, “using the most significant and/or frequent codes” to create rough categories (Charmaz, 2014, pp. 57-58). These categories can be related to each other and, if needed, subcategories can be established. (Charmaz, 2014). Throughout these phases, memo-writing takes place, which is writing short anecdotal notes about any thoughts that come up during the coding of data to spark reflections (Charmaz, 2014). To fill in gaps in the data, through theoretical sampling, new data and coding can be conducted (Charmaz, 2014). Working iteratively, this process is repeated with new data and coding, until no new insights are sparked, which Charmaz (2014) describes as saturation. To finalize the process, core categories are identified and a theory is constructed, which should represent the data, categories, and their relationships (Charmaz, 2014).

Grounded theory is particularly well suited to be applied in the project, due to its multiple strengths, specific to the project’s characteristics. AI in art is a relatively new phenomenon, while grounded theory offers a systematic approach to navigating novel research topics. Therefore, grounded theory is a useful and promising method for the project. Additionally, grounded theory is flexible regarding what kind of data can be used, which is valuable, because the project incorporates generative workshops, exploratory workshops, interviews, and field research. Moreover, the iterative nature of grounded theory, continuing cycles of data collection, coding, and analysis, aligns with the mentioned multiple activities of the design process and its way of developing the project. Furthermore, since the project applies PD, the emphasis on the participants’ perspective present in grounded theory fits with the project’s methodological approach.

Due to the scope of the project, I applied the method of grounded theory in a limited way, as applying the method extensively to its detail would not have been a productive use of my time. My initial coding was not conducted word-by-word but in a more general sense. Additionally, even though I conducted multiple activities which produced data, this cannot be seen as theoretical sampling, since these activities were not primarily intended to fill in specific gaps in the data. Therefore, I do not claim to have reached saturation and did not practice memo-writing. However, this does not diminish the meaningfulness of the resulting data and categories. Furthermore, I did not formulate one final theory, as this is represented in my design proposal.

4 Field Research

4.1 Method

4.1.1 Design Ethnography

The method of design ethnography is the study of groups of people by observing and participating in their lives to provide rich insights into their culture and surroundings (Muratovski, 2021). A variety of methods fall under design ethnography. More specifically, I practised participant observation, which can be described as experiencing a culture by participating in an activity to gain an understanding of its people and their behaviour (Hanington & Martin, 2012).

Conducting ethnographic design research, I went to multiple galleries, museums, and art schools in Malmö to participate in activities, observe people, and join the discourse on AI in art.

4.1.2 Semi-Structured Interviews

Semi-structured interviews were conducted to gain insight into the personal experiences, opinions, and perspectives of people relevant to the project. They combine predetermined questions on a set of topics with an open exploration of interviewees' answers by allowing for improvised follow-up questions on emerging themes (Wilson, 2014). They were used, because of their strength to allow for exploration and open-ended discussion. An interview guide, a document outlining the structure of the interview, listing questions and topics to focus on, and containing suggested spontaneous prompts, was used during the interviews.

As part of the initial research phase, I conducted seven semi-structured interviews. Involving a diverse set of creatives, interviewees' ages ranged from their mid-twenties to mid-sixties. Two women and five men were interviewed. All participants were based in Scania, Sweden at the time of interviewing, but their cultural backgrounds varied, having grown up in Sweden, the USA, France, Poland, and Germany. Three interviews were conducted in person and four interviews were conducted remotely. They lasted about 30 to 40 minutes. The interviewees were recruited through personal connections, networking at local events for creatives, and through a call for participation, shared on social media. Through audio recording, the interviews were documented and transcribed.

Following the interview guide, the semi-structured interviews were split into five phases: (1) an introduction of the project including an explanation of the purpose of the interview, (2) questions about the interviewee's background as an artist, (3) questions about the interviewee's view of and experience with generative AI, (4) questions about their perspective on generative AI in art,

and (5) questions about the interviewee's view on the implications of generative AI for society.

Questions asked in phase 3 included, but were not limited to, "*How do you see recent developments in generative AI?*", "*Have you used generative AI tools, and if so how?*", and "*How does generative AI affect you as an artist?*".

In phase 4, I asked questions such as, "*Do you think AI-generated material can be considered as art, if so why or why not?*", "*What constitutes art?*", and "*Would you use generative AI in your own creative practice, if so why or why not?*".

Furthermore, in phase 5, questions such as "*In your opinion, what are the implications of widespread use of generative AI for society?*" and "*How will the use of generative AI affect society's perception of art*" were asked.

Interviewee 1 draws digitally and traditionally in their free time and chose the art specialisation during high school, currently studying at university. Interviewee 2 has been freelancing as a digital artist for seven years, drawing characters and environments for games and related entertainment. Interviewee 3 draws, paints, and has completed their education at a formal art school, currently being unemployed. As a musician, interviewee 4 takes gigs as a DJ and considers themselves spiritual. Interviewee 5 is an art director and animator, working with visual graphics. As a fine artist, Interviewee 6 exhibited at galleries in Sweden and Europe, and they are a consultant to a governmental art agency. Lastly, Interviewee 7 is an art historian, a lecturer, and a curator of a prominent museum in Scania.

4.2 Results

4.2.1 Design Ethnography

To gain an understanding of how artists see the issue of AI in art, I participated in a workshop at the Inter Arts Center, Lund University, in Malmö called "Beyond hallucinations & dreams: Experimental approaches to generative AI models". I gained a perspective on how AI is addressed in artistic practice and how artists aim to use these AI tools.

Furthermore, I established connections at the Cowgirl Gallery in Malmö and informally talked to people about their views of AI. Many artists who frequent the gallery exhibited an outspoken disregard for recent developments in generative AI.

4.2.2 Semi-Structured Interviews

After initial and focused coding of the interview transcriptions, five major themes emerged: (1) the practice of using artists' work in AI training without consent, (2) the uncertainty of employment for artists, (3) the use of generative AI tools within the artistic process, (4) the difference between

human art and AI-generated art, and (5) the view of a future which is oversaturated by AI art.

Employment

The Interviewees' feelings towards the uncertainty of employment of artists due to AI ranged from personal anxiety about their own professional prospects to expressing that their own employment would be safe, due to a lack of quality of AI-generated material.

Interviewee 3 expressed their concern about finding employment in the future, as follows:

> *At first, it was very discouraging of course, because it's like, well I'm never going to get a job now. And to some degree, I still feel that way, because I know a lot of people I know have been replaced by AI in the industry. They have been, you know, just fired.*

Additionally, interviewees saw mainly junior roles in artistic professions impacted, because their tasks could easily be automated. This could have future implications, as no new artists with senior experience follow, as interviewee 1 pointed out.

Nonetheless, most of the interviewees felt confident that their employment would not be impacted by AI. They expressed that the quality of AI-generated material is not sufficient, especially when it comes to details. Generative AI is only good for quantity and not quality, as they stated. Yet, some interviewees speculated that employers could disregard quality, laying off artists in favour of AI use to increase profits.

AI Training Practice

Interviewees expressed a range of views on the practice of training generative AI on artists' work without consent, ranging from seeing it as entirely unacceptable to comparing it to common artistic inspiration, while the majority was somewhere in between the two positions.

Some interviewees emphasised that they see the use of artists' works in AI training without consent as fundamentally unethical. Interviewee 2 states that "the big concern is that it learns off of peoples' art without permission". Interviewee 1 mentioned the following, "Yeah, and doing this [training practice], I think, is highly unethical, uh, because these artists are not seeing any compensation, but actively having their work being stolen to generate new things, without their consent". What is more, Interviewees argued that the training practice is unfair, because resulting AI models are competing with artists.

However, some interviewees expressed that training AI on copyrighted material is not the main problem. The training practice was seen to be comparable to artistic inspiration as practised commonly between artists,

which is acceptable, as long as AI tools are not used to precisely copy someone's artistic style or to try to generate profits. Yet, interviewees recognised that the practice of training AI on artists work is much more direct than mentioned general artistic inspiration. Among interviewees, there was a sense of artists being in danger in the current society, besides the training practice, for example, due to a cut of funding for culture. Thus, interviewees called for a kind of protection for artists.

Nonetheless, interviewee 4 saw human artistic inspiration and the training of AI on artists' works as entirely equal, stating that "we all just reinterpret".

AI Tools in the Artistic Process

Regarding the use of AI in the artistic process, interviewees' attitudes reached from not accepting any AI in the artistic process, over being open to the possibility, to freely using it in their artistic process. Most interviewees were open to the idea of automating minor tasks in the artistic process using an AI tool.

Interviewee 3 expressed that they do not use any AI tools in their process by stating the following, "I would feel that it is fake. I like to have complete creative control of what I do. You know? And you don't have that with AI." Additionally, the unethical training practice was mentioned by interviewees as a reason to not use AI tools at all.

Others see potential for the use of AI tools in the artistic process, even while not currently using any. They see a use case mainly for automating smaller tedious tasks, for example generating textures for otherwise handmade 3D models. Interviewees mention openness to train their own AI models on their own art. They emphasise that this is based on the choice of the artist.

Expressing an enthusiastic view, interviewee 4 uses AI tools to get inspired during their artistic process. They talk out loud to an AI chat tool, which visualises part of their conversation through an AI image generation tool. They stated that this practice of visualising their ideas through AI helps them get inspired.

Human Art and AI-generated Art

All interviewees expressed a strong belief that there is a human quality in art that AI tools cannot reach, except one interviewee who was of a different opinion.

Interviewees stated that AI art can easily be identified as such and that it feels flat and soulless. Additionally, many interviewees referred to a human quality in art, which they said is unobtainable to AI. Interviewee 6 emphasized the role of the body in her artistic process. They stated the following, "For me [creating art] is so physical. We are bodies, intelligent bodies, [...] and I don't think AI can reach that." Other interviewees stated that the human quality in art is about personal experience, human emotion, or imperfections.

Interviewee 4 expressed the view that there is no uniquely human quality in art. They stated the following, “It is just about artists’ ego, you know? And they feel so threatened by anything that places them lower in, like, their own system of standards.”

Furthermore, in connection to these questions, interviewees were asked to define art. While some interviewees were open about what can be art, others had a more restricted view. Interviewee 7 stated that “art is about having something to say”, while Interviewee 6 noted that art is about going deep into a topic as a way of searching for complexity. Interviewee 6 added the following, “Already 99% of the population have no idea about what art is about.”. Others had a different opinion. Interviewee 3 stated the following, “I used to think that art needs to be something, but, like, now I think art can be anything”. Many interviewees felt that identifying something as art depends on the perspective of the viewer.

AI Art in the Future

All interviewees had concerns regarding the impact of AI tools on the future. They described their view of a future, which is oversaturated by AI art, having varying consequences. Some interviewees mentioned that they could see human art being valued more by society, while others thought art as a whole, including AI-generated art and human art, would lose value, as people get tired of large quantities of AI-generated art. Another interviewee saw a future in which artists make their art less accessible to avoid it from being used as training data, potentially limiting access to physical exhibitions only.

As a way of preventing such developments, interviewees placed their hope on legislation. They mentioned copyright lawsuits as potentially being able to have an impact, or they stated that general legislation, potentially by the EU, could protect artists’ rights to their work, for example, to be excluded from training data. Additionally, an interviewee mentioned that increased art education could spread more awareness of what art is about, but added that such education is already being defunded.

4.3 Discussion

Addressing one of the sub-questions of my research, “How do creatives see AI in relation to art and society and what are the implications?”, these interviews show a diverse range of views among artists of AI in relation to art and society. It can be said that the opinion of the interviewees leans strongly towards a critical view of AI, with exceptions, however. Mainly, interviewee 4 expressed an otherwise unseen enthusiasm about AI and its use in art. Otherwise, different attitudes towards art can be seen among interviewees. Having a background in fine art, interviewees 6 and 7 have a conservative view of what art can be, while the other interviewees expressed a more liberal view.

Applying the method of grounded theory the five themes outlined above represent categories that emerged during the coding of the interview data, partly having subcategories or showing relation to each other. These categories will be expanded upon with additional data, coming from the other activities.

Furthermore, the second part of the sub-question, “What are the implications?”, remains to be addressed. Therefore, the following workshop aims to facilitate discussion among artists about the implied futures of AI and art. The workshop will gather ideas on current developments and trends in the field of AI and use those to ideate on possible futures. Furthermore, the workshop focuses more on collaboration and a shared discussion rather than isolated interviews.

5 Workshop: Artists on AI in Art and Society

5.1 Method

5.1.1 Design Workshop

Often used in PD design workshops entail multiple Co-Design activities to creatively engage participants (Hanington & Martin, 2012). Their strength lies in collecting a wide range of insights from participants and they can be used during exploratory and generative phases of the design process (Hanington & Martin, 2012).

The goal of this workshop was to create a discourse among artists about AI in relation to art and society as a form of inquiry. During the workshop, the emphasis was placed on past, present and future influences on AI from the perspective of artists. Additionally, ideas on probable and preferable futures were gathered to inform the next speculative workshop together with the data from the field research and interviews. The concepts of the futures triangle, as used by Inayatullah (2023), and the futures cone, by Taylor (1990), were utilized in the workshop, touching upon the field of futures studies or foresight studies.

To enable discussion about the futures of AI in art, the futures triangle as first mentioned by Inayatullah and Wildman (1998), was used. As seen in Figure 1, it maps three factors for potential futures, which are the push of the present, the pull of the future, and the weight of the past (Inayatullah, 2023). The push of the present describes developments and trends, already in motion, that are shaping the future. The pull from the future represents a

novelty or a signal of change, and the weight of the past is an influential societal factor or structure which resists change, upholding the status quo.

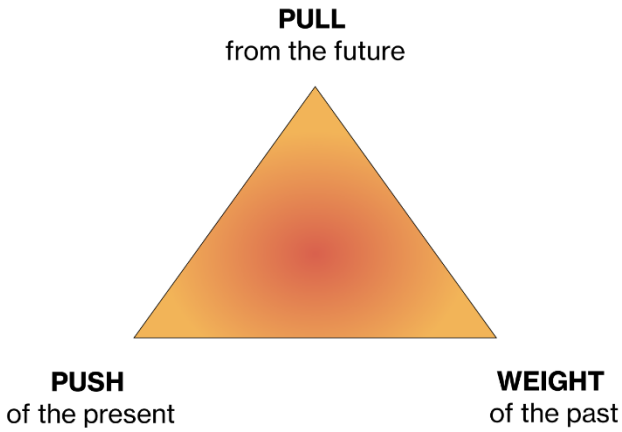


Figure 1. The Futures Triangle

To facilitate futures thinking among the participants, the workshop made use of the futures cone, first used by Taylor (1990) according to Voros (2017). The futures cone, as a model, illustrates the idea of multiple futures expanding in possibilities over time (Hancock & Bezold, 1994; Taylor, 1990; von Hippel, 2005). As seen in Figure 2, it differentiates between probable, preferable, plausible, and possible futures. It has prominently been used by Dunne and Raby (2013) to contextualize their SD approach by establishing the idea of multiple futures.

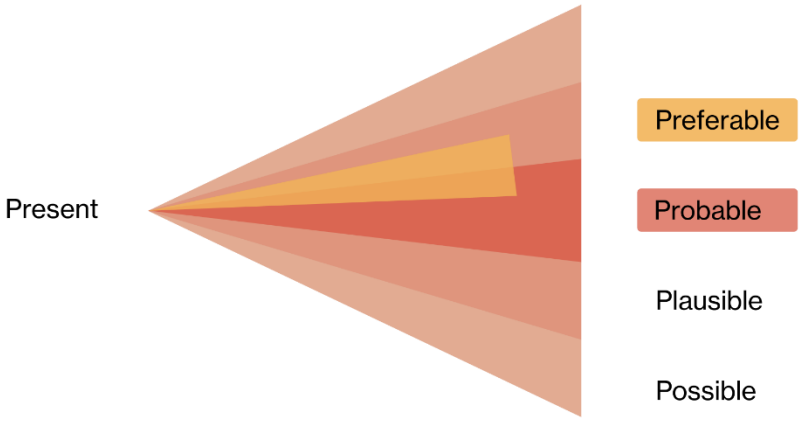


Figure 2. The Futures Cone

Participants

To recruit a range of artists with different backgrounds as participants in the workshop, the call for participation was shared across various networks and locations. Throughout Malmö, posters, as seen in Figure 3, were put up at art galleries, museums, music schools, art schools, university buildings and

libraries. Using a Facebook event page, the workshop was distributed online by inviting relevant people. Additionally, people who had been engaged in the project in fieldwork or interviews were invited directly to the workshop. Furthermore, no registration was required to attend, to make the workshop openly available to the public.

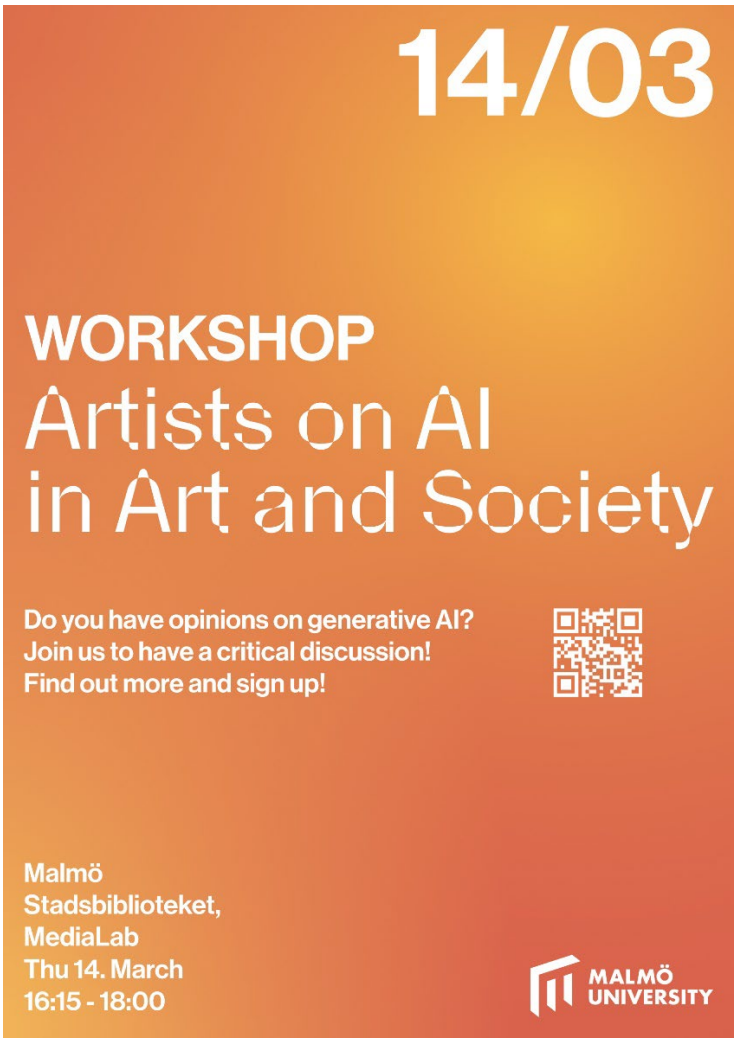


Figure 3. Poster for the Artists on AI in Art and Society Workshop

The workshop included five participants, three women and two men. Three participants were in their twenties, one was in their thirties, and one was in their forties. This information is provided to document the age range of the participants. Participant 1 was a fine artist with a traditional art education, primarily practising on canvases using charcoal and collages. Participant 2 was a software engineer, working with computer vision, and drawing in their free time. Participant 3 was an interaction designer, interested in AI and its creative uses. Participant 4 was an artist casually drawing digitally and traditionally, while also being an interaction designer. Finally, Participant 5 was a musician and interaction designer.

Workshop Structure

The workshop was split into three parts: (1) an introduction, (2) a phase dedicated to working with the futures triangle, and (3) a phase for ideating on futures using the futures cone. The workshop was guided by a presentation, which can be found online[1].

Beginning the workshop, there was a short introduction to the project and the workshop. Participants were informed that the materials they produced will be used in this thesis project and that, if they agree to it, pictures will be taken. Previous insights from the fieldwork and interviews were presented to the participants. Next, participants were asked to introduce themselves and their backgrounds as creatives, and to state what had intrigued them to join the workshop.

The second part of the workshop started by explaining the concept of the futures triangle, as used by Inayatullah (2023). A whiteboard was set up with the futures triangle and participants were provided with sticky notes. Of the three corners, each was presented one by one, using examples within the topic of generative AI. Five minutes for each of the corners were given to the participants to write down their ideas on sticky notes and place them on the whiteboard. Once this process was completed for all three corners, the participants were encouraged to take a closer look at the final futures triangle with ideas placed around it, to discuss what caught their attention.

Going into the next phase, the futures cone was briefly explained, focusing on probable and preferable futures. For this task, participants placed sticky notes with ideas for futures on a whiteboard around the two types of futures. After familiarising all participants with the concepts, they were given five minutes to ideate on probable futures of AI, write them on sticky notes, and place them on the whiteboard. The previous task should be used as inspiration. The emphasis was on the next task of ideating for ten minutes about preferable futures of AI and placing them on the whiteboard as well. Being imaginative was stressed here to gather creative ideas. This was followed by a discussion about the ideas and their interesting qualities.

To conclude the workshop, participants were asked to briefly talk about their experience ideating about the future of AI and what they had found intriguing. Finally, information about the next workshop was presented encouraging participants to attend.

Materials and Tools

The workshop materials were mainly sticky notes and pens, to write down ideas and place them on whiteboards depending on the task.

---

[1] Link to the presentation:
https://drive.google.com/file/d/1OGAWg8lxG3gQPQGrWaQNUs9UBRQ32vVs/

Data Collection

During the workshop, data collection included observations and gathering notes about the discussions between participants. The resulting sticky notes on the whiteboards were documented in pictures.

Ethical Considerations

All participants were provided with information about the project, the workshop, and their rights. Informed consent was obtained from each participant prior to the workshop to use the produced material and to take pictures during the workshop. The collected material did not contain any personal or sensitive information.

5.2 Results

The workshop unfolded as stated previously with an introduction, a phase dedicated to working with the futures triangle, as seen in Figure 4, and a phase for ideating on futures using the futures cone. During and between the tasks, participants were engaged in discussions about the relevant topics, as seen in Figure 5.



Figure 4. Workshop participants work with the Futures Triangle

5.2.1 Futures Triangle Task

Working with the Futures Triangle, Participants contributed a variety of ideas to each of the corners. The final model with the participants' sticky notes can be seen in Figure 6.



Figure 5. Workshop participants engage in discussion

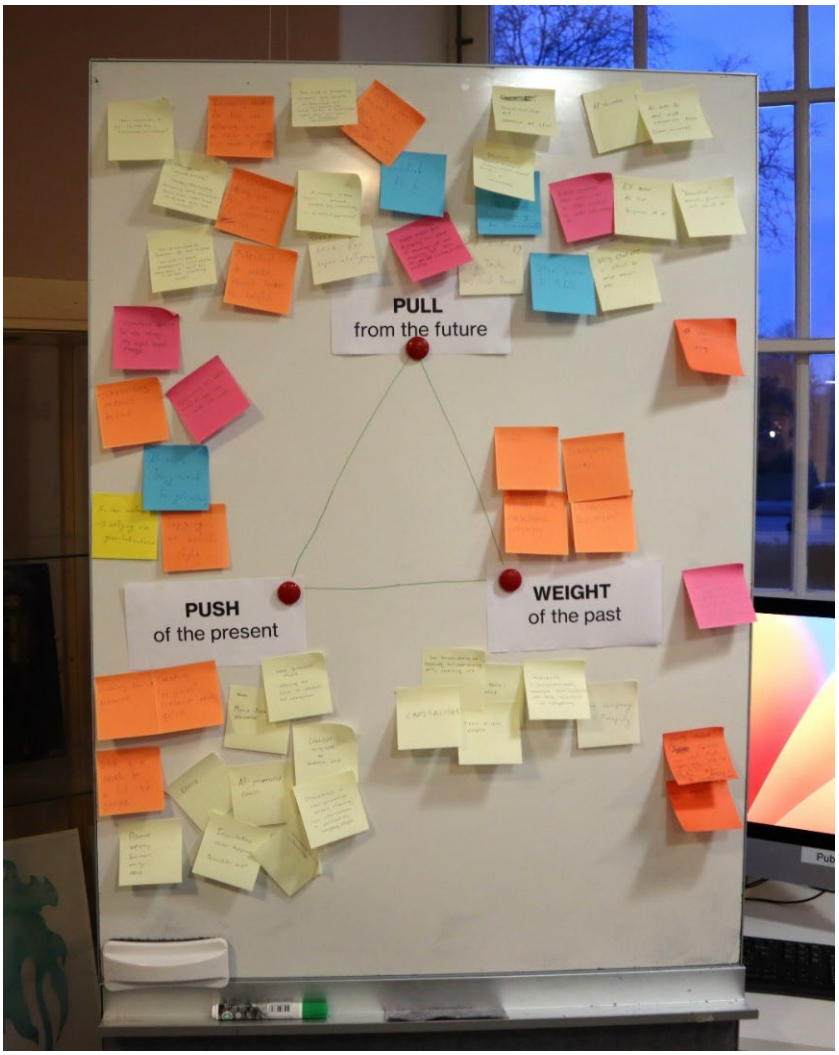


Figure 6. The Futures Triangle with sticky notes

The Push of the Present

Participants wrote down a variety of developments and trends for the push of the present. They noted that AI is used to automate a variety of tasks, such as coding, translation, and even academic work. Furthermore, the creative use of AI was very present in the participants' views of current developments. This included mentions of AI-generated images and poetry, noting that artists' styles are being copied. One note stated that there is the development of a constant need for new things, establishing a rapid pace of media development. A low bar of entry to create AI images was noted, and the integration of AI into tools like Photoshop was referred to. Additionally, some participants noted developments such as a declining trust in information, due to deep fakes and image generation.

The Pull from the Future

Multiple ideas on novelties or signals of change were noted by the participants regarding the pull from the future. Regarding art, this included a change in how art is defined, caused by new developments of AI, and potentially new art mediums. Participants mentioned the novelty of utilizing AI in artistic processes, such as a musician making their voice available to use for other artists, by training their own AI model. On a more general level, participants noted that technology is becoming more accessible, that virtual reality is developing, that tasks can be automated by AI, and that AI tools are being utilized in things like customer service. Furthermore, on the topic of ownership, open-source AI models and new legislation by the EU were mentioned. Additionally, new research beyond humans, focusing on animals, plants, and nonhuman beings, like AI, was noted.

The Weight of the Past

Regarding the weight of the past, which is an influential societal factor that resists change, participants noted the following. Profit and big companies monopolizing technology were the major themes for participants. Additionally, transparency issues, privacy and intellectual property were mentioned to potentially resist change. Furthermore, participants noted a fear of change by the people as a limiting factor for change.

5.2.2 Futures Cone Task

Working with the Futures Cone, Participants noted down ideas on probable and preferable futures of generative AI in art using sticky notes, as seen in Figure 7.

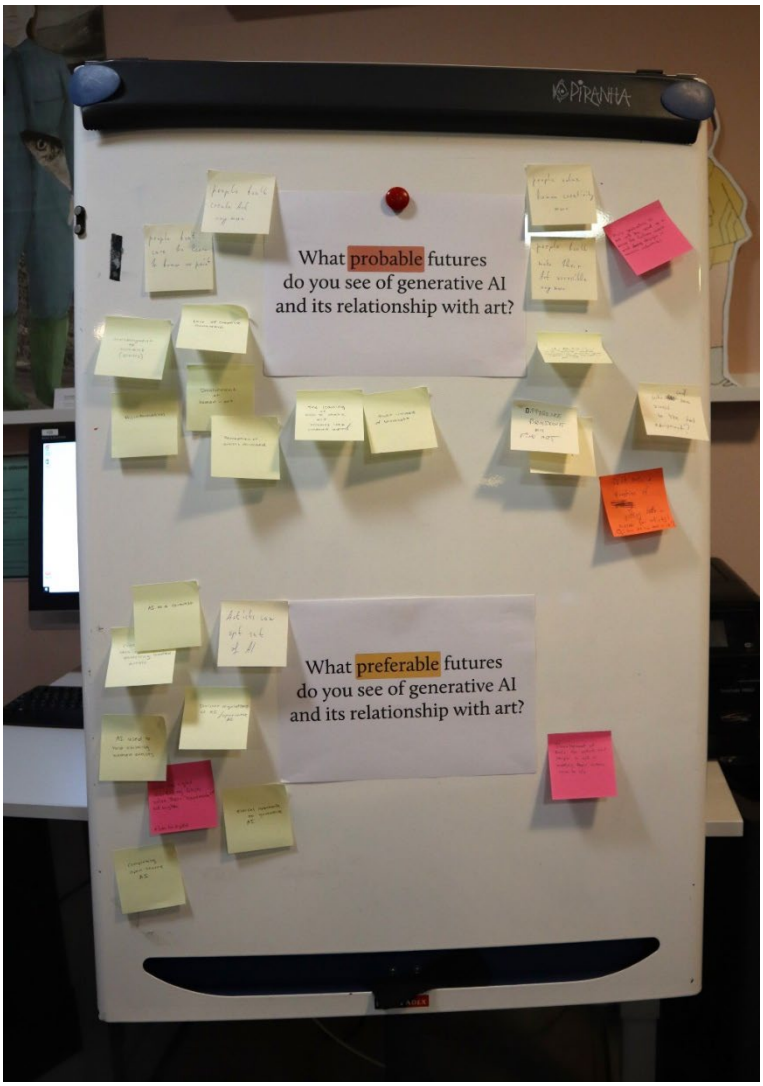


Figure 7. Sticky Notes for Probable and Preferable Futures

Probable Futures

Informed by the previous task of working with the Futures Triangle, participants wrote down ideas on what kind of probable future follows from the previously discussed influences. A central theme was that artistic practices like drawing and painting would be lost, as people did not care to learn such skills anymore. Thus, there is a devaluation of art and artists face unemployment. A participant noted that people would not make their art accessible anymore and that there would be a loss of creative ownership. Furthermore, participants mentioned a profit increase for big tech companies and an increase in misinformation.

Preferable Futures

Regarding the more imaginative task of ideating on preferable futures, participants mentioned open-source AI models, ethical AI, and the option for artists to opt out of training. Additionally, there were mentions of AI tools for artists as an aid to making their visions come to life. The idea of AI as a co-

artist was noted. Furthermore, stricter regulations and laws that protect artists came up, as well as ideas of a future where handmade art is valued more.

5.3 Discussion

The workshop provided further data to address the research sub-question, "How do creatives see AI in relation to art and society and what are the implications?", through the Future Triangle task and the corresponding discussion. Additionally, the workshop participants explored the implications of their views, in the Futures Cone task, by ideating on probable and preferable futures. What is more, the collaborative workshop format contributed to exploring the second sub-question of my research, addressing the collaborative speculation approach.

Reflecting on the workshop, the focus on AI in art rather than AI in general might not have been entirely clear to all participants during the different tasks, since the notes also address issues unrelated to art as such. Therefore, I should have articulated this focus more clearly. Furthermore, addressing the Futures Cone task, the preferable futures ideas are not as imaginative as I hoped for. Even though the speculative intent of the task was emphasised during the workshop, giving an example could have communicated this more clearly.

Continuing the method of grounded theory, the relevant data from the workshop was coded and sorted to the five themes that were present from the field research, adding further depth to the data. From these themes, "What if?" questions were prepared for the following speculation workshop, which can be seen as the main speculative activity of the project, addressing the research question "How might we speculate through design about the relationship between AI, art, and creatives in society in the present and future?", as well as both sub-questions.

6 Workshop: Speculative Futures of AI in Art

6.1 Method

6.1.1 Design Fiction

As a SD method, design fiction aims to communicate possible futures, often through scenarios and storytelling (Markussen & Knutz, 2013). A common way of starting the imagination process is through "What if?" questions (Dunne & Raby, 2013), as applied in this workshop. After having imagined

fictional worlds through scenarios considering technological, social, economic, and environmental factors, the next step is often to imagine fictional artefacts that embody said factors (Dunne & Raby, 2013). Through prototyping, these artefacts can then be further developed.

6.1.2 Design Workshop

The goal of this workshop was to enable artists to collaboratively imagine speculative futures of AI in art to spark reflection, critique, provocation, and inspiration. The participants created scenarios of futures that might or might not be preferable and, within those scenarios, they designed fictional objects that embody certain values. The emphasis is on being imaginative to create a discussion. The material from the previous fieldwork, interviews, and workshop informed the starting point of this workshop in the form of "What if?" questions. The seven questions were picked to facilitate imaginative and radical thinking, therefore themes such as fears of unemployment were left out. The questions can be seen below.

1. *What if AI was integrated into physical objects in our environment (furniture, buildings, etc.), making its presence constantly felt, yet less identifiable?*
2. *What if generative AI was a widely used tool within the artistic process, comparable to a pen, an instrument or Photoshop?*
3. *What if AI models could dynamically inspire and stimulate artists in their process, similar to a muse?*
4. *What if completely new art mediums emerged, because of generative AI?*
5. *What if generative AI was fully integrated into every part of our mundane daily life?*
6. *What if there were publicly owned generative AI models?*
7. *What if humans and generative AI developed as co-artists where the line between creator and assistant becomes indistinguishable?*

After the workshop, the resulting ideas, imagined by participants, were iterated on to refine the scenarios and artefacts into short stories with visualisations in the next step of the project.

Participants

A main priority in preparation for the workshop was to engage a diverse group of participants in the workshop. Therefore, the call for participation was distributed across a diverse range of locations and networks to reach artists and creatives in various domains. Posters, as seen in Figure 8, were put up at numerous locations in Malmö, including art galleries, museums, music schools, art schools, university buildings, and cafes. A Facebook event page was used to distribute information online, by sending it to people who had been engaged in the project in fieldwork, interviews or in the previous workshop. Additionally, the workshop was posted in a local newsletter

informing about current events in Malmö. Furthermore, the workshop was part of a series of lectures and events on AI at the Malmö City Library, called “Nyfiken på AI”, which was promoted around the city. Furthermore, while mainly targeting artists, the workshop was open to the public to ensure accessibility. Therefore, registration was not mandatory.



Figure 8. Poster for the Speculative Futures of AI in Art Workshop

The workshop included eight participants, with four women and four men. The group comprised artists and creatives from various backgrounds. Four participants were in their twenties. Two participants were in their thirties and the other two were in their fifties, which is relevant, to prove a diverse age range. Additionally, age might give an indication of participants’ familiarity with, and attitude towards, technology and AI. Furthermore, one person who had attended the previous workshop participated, while no interviewees joined the workshop. Participant 1 is a documentary filmmaker, who is curious to use AI in their work. Participant 2 enjoys making music and drawing in their free time. Participant 3 is a senior digital designer, has

worked with machine learning, and is interested in the collaborative workshop format. Participant 4 studies visual culture and has worked in architecture. Participant 5 is broadly interested in digital art and AI. Participant 6 is a municipality worker, who primarily works with children and has recently been engaged in teaching children about new technologies and AI. Participant 7 is a digital artist who draws digitally and physically and is an interaction designer. Participant 8 is a musician and an interaction designer.

Workshop Structure

The workshop was split into three parts: (1) an introduction, (2) a phase dedicated to imaging and building speculative scenarios, and (3) a phase focused on designing fictional objects. The workshop was aided by a presentation, which can be found online[2].

Starting off, there was a brief introduction of the project, its purpose, and the programme of the workshop. Participants were informed that the materials they produced will be used in this thesis project and that, if they agree to it, pictures will be taken. The insights from the fieldwork, the interviews, and the previous workshop were presented in the form of relevant themes. A personal introduction round followed. Participants were asked to introduce themselves and their creative and professional backgrounds. Additionally, they were asked to state what had intrigued them to join the workshop. When each participant had introduced themselves, a short explanation of SD was presented to conclude the introduction phase of the workshop.

Next, the concrete task of scenario building was introduced, which was to brainstorm a future scenario inspired by a "what if?" question. The scenario should be set in the year 2050, involve one or more characters who are artists, and be set in, but not limited to, the artist's studio. Any kind of artist may be chosen, for example, visual artists, musicians, performance artists or filmmakers. Four spontaneous groups of two participants were formed and a document with links for generative AI tools was provided for optional use during the task. The "What if?" questions were given to the participants to choose from, along with worksheets to document their ideas. Here, the groupwork to imagine the scenarios started. A time period of roughly 20 minutes was reserved for the task. Participants could ask questions to clarify the task or the provided material. After the work period was over, the participants presented their scenario idea to the group and gave each other feedback.

After a short break, the next phase of the workshop began, which expanded on the previously imagined scenario. The task was to create a fictional object that exists within the scenario. Participants should ideate on problems and

[2] Link to the presentation:
https://drive.google.com/file/d/1jwDbPcTvV8NY-uHFVobJ6Cp4YhrW8abu/

challenges in their scenario to identify new needs and wants which could be addressed by their object. They should try to visualize the object through sketches and come up with a name for it. Worksheets were provided for the task, which was planned to take about 20 minutes. As the time period was over, the participants again presented their designs to the group and gave each other feedback.

To conclude the workshop, participants were asked to reflect on the workshop, tell the group what they found interesting about the scenarios, and share what they learned. Finally, the next steps in the project were presented to the participants to inform them of what happens with the produced materials.

Materials and Tools

Various materials were utilized throughout the workshop, including sticky notes, coloured pens, and paper to facilitate idea generation. Physical copies of the “What if?” questions were handed out, to enable the discussion omg groups about choosing one of them. Furthermore, two worksheets were provided to guide the participants during the tasks and to document their ideas. For the scenario task, the worksheet, as seen in Figure 9, had three fields, one for world building, one for characters, and a text field to write out the scenario. For the design task, the worksheet, as seen in Figure 10, had three fields, one main field for sketches, one for a description of the fictional object, and one to explain how it fits into the scenario.

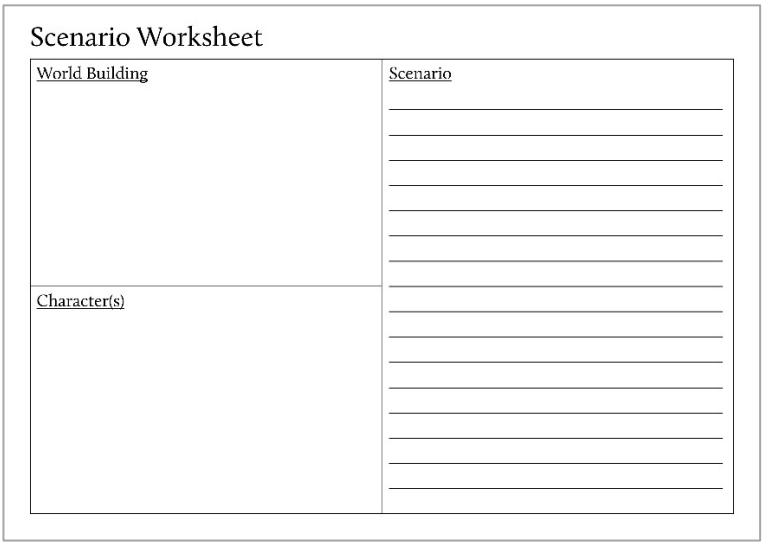


Figure 9. Worksheet for Scenario Task

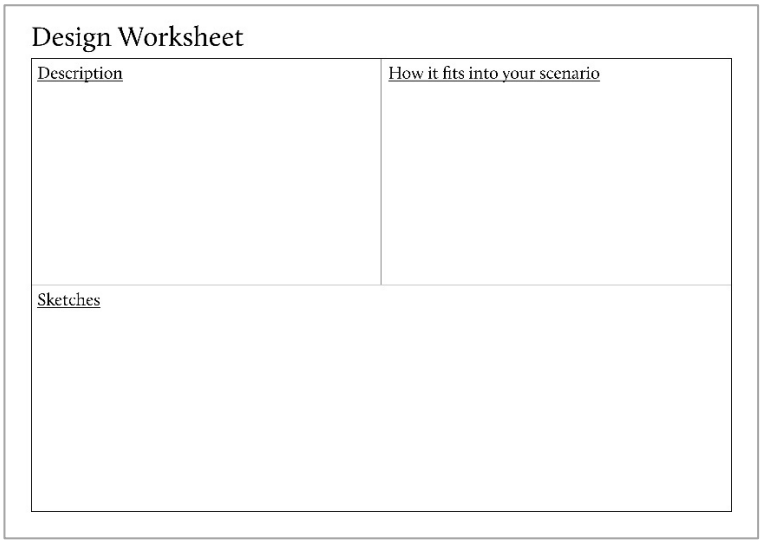


Figure 10. Worksheet for Design Task

Regarding digital tools, a document with links to generative AI tools for optional use during the tasks was provided. The document included text, image, video, sound, and 3D generation tools and can be found online[3].

Data Collection

During the workshop, data collection included observation and gathering material and notes. Observations were recorded in the form of notes during the workshop, focusing on the open-ended discussions at the end of each task.

Ethical Considerations

All participants were provided with information about the project, the workshop, and their rights. Informed consent was obtained from each participant prior to the workshop to use the produced material and to take pictures during the workshop. The collected material did not contain any personal or sensitive information.

6.2 Results

The workshop followed the structure outlined above, with the exception that during both tasks more time was given, around 30 minutes, because the groups expressed a need for more time, and other parts of the workshop took less time than expected. The workshop set up with engaged participants can be seen in Figure 11 and Figure 12. The provided material, as described previously, can be seen in Figure 13 and Figure 14. One of the eight participants was open to experimenting with generative AI during the workshop tasks, using a variety of AI models, including ChatGPT for text generation and ModelScope Text-to-Video for video generation.

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[3] Link to the AI resources document:
https://drive.google.com/file/d/15h-AhqajzuotsnAvJ5L1ktaom-8iL5G2/



Figure 11. Workshop Setup (1)



Figure 12. Workshop Setup (2)



Figure 13. Workshop Materials in Action

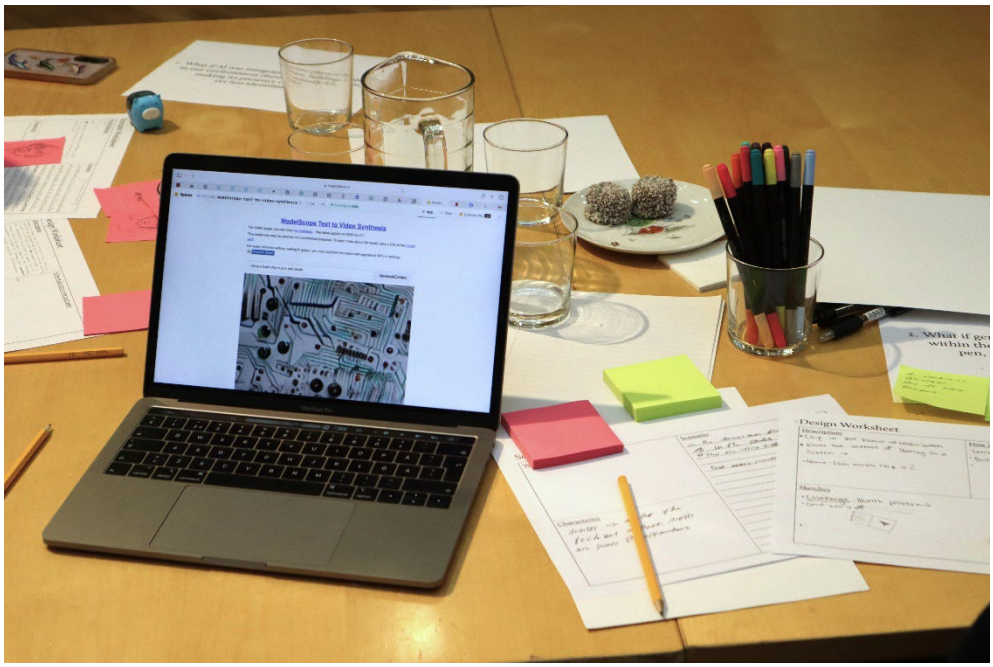


Figure 14. Workshop Materials with a focus on the use of an AI tool

6.2.1 Group 1: The AI-Muse

Group 1 decided to work with the following question, “What if AI models could dynamically inspire and stimulate artists in their process, similar to a muse?”. The produced material can be seen in Figure 15.

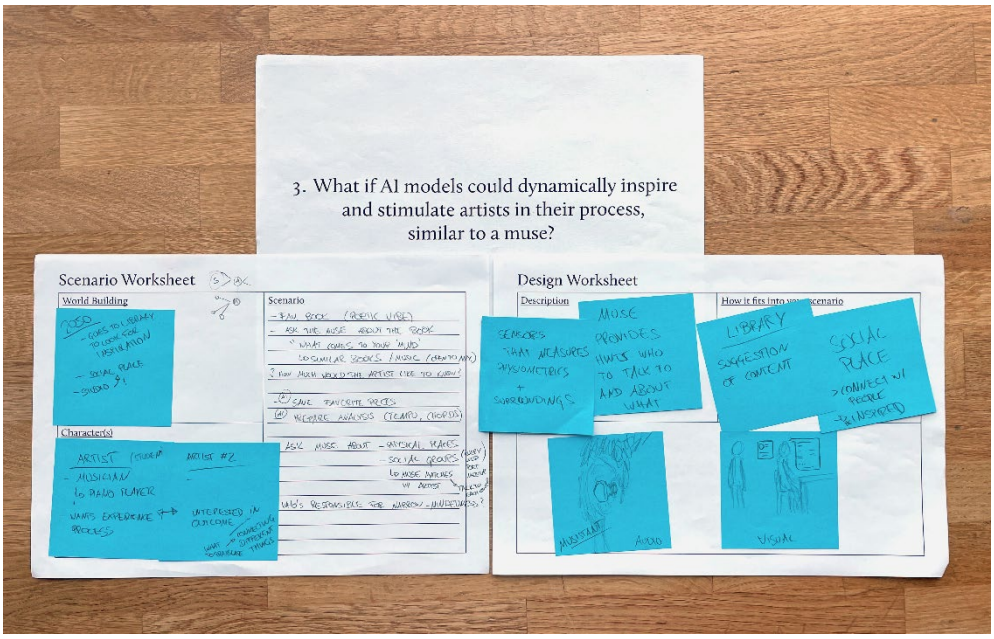


Figure 15. Material of Group 1

Based on the workshop material, the presentation, and the discussion during the workshop, group 1 came up with a scenario that is set in a library. The main character is an artist, who is a piano player and is looking for inspiration. Looking for his favourite book, he asks the AI muse for similar material. The AI muse provides chords and tempo based on his book. Additionally, the AI muse provides information about social places and meetups at the library. It matches the main character with other artists who are also using an AI muse. He talks to another artist about his artistic process with the muse. He then proceeds to his studio.

Focusing on the design of the AI muse device, it is an earpiece that talks to the user. The device has sensors that measure physio metrics and the surroundings. The group provided sketches of the device and how it would provide visual information.

6.2.2 Group 2: The Creature Generator

Group 2 chose to start the tasks with the following question, “What if completely new art mediums emerged, because of generative AI?” and their workshop material can be seen in Figure 16.

Group 2 imagined a scenario that is set in a rural world where people spend their lives in virtual reality (VR). The main character is called “Spark 27” and is a performance artist, who is working from a wooden den. For his art, he creates new types of animals, fantastic beasts, using generative AI by combining the DNA of animals that are almost extinct. These creatures exist in the VR world and people can interact with the animals or even experience the world through their eyes.

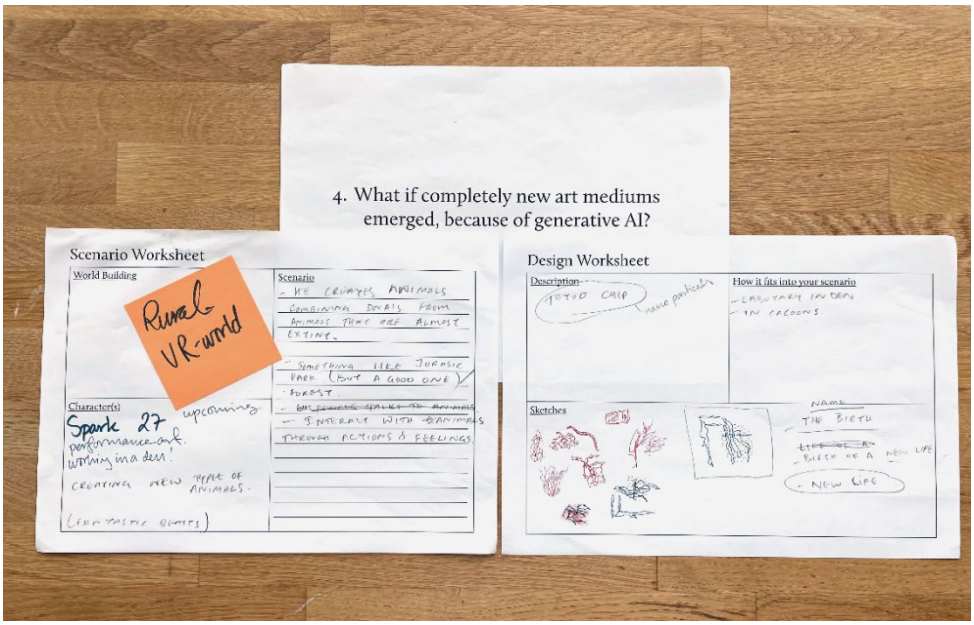


Figure 16. Material of Group 2

In this scenario, the group designed a creature generator which is a computer chip named “New Life”. It is located in the main character’s laboratory in his wooden den. Furthermore, the group sketched out some creatures that could be created.

6.2.3 Group 3: The AI-Scanner

Group 3 picked the following question as a starting point for their scenario, “What if AI was integrated into physical objects in our environment (furniture, buildings, etc), making its presence constantly felt, yet less identifiable?”. Their workshop material can be seen in Figure 17.

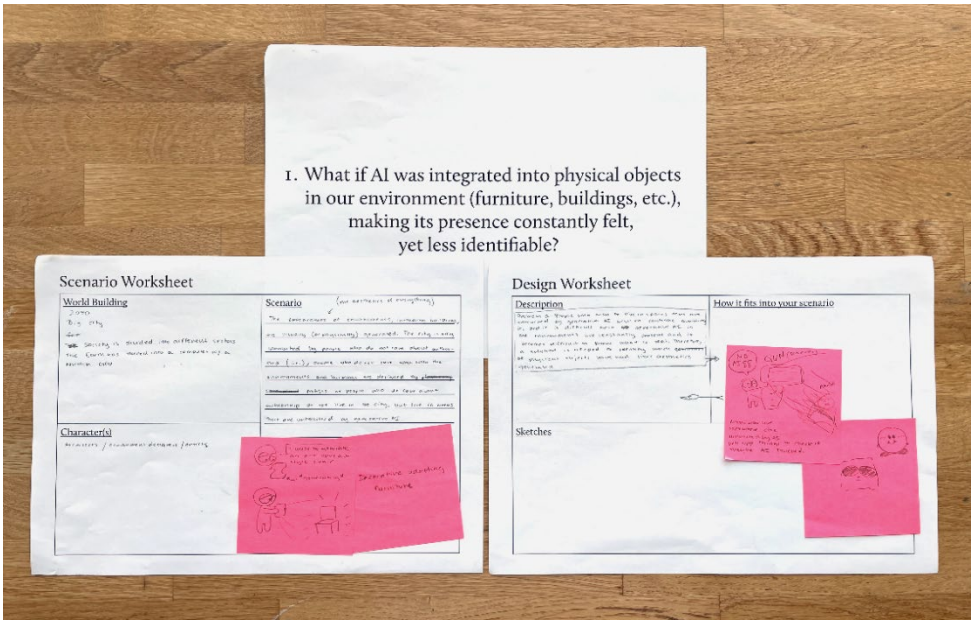


Figure 17. Material of Group 3

Group 3 imagined a scenario which is set in a society that is divided into two parts. In the big cities, the appearance of most spaces and buildings is AI-generated. People do not care who the environments are designed by. Artists and people who care about authorship do not live in the cities, but live in areas that are untouched by generative AI.

The group designed a device that can detect whether any object is AI-generated. This is needed, because artists who live away from the AI-dominated cities face the problem of identifying AI-generated things. The scanner device is shaped like a gun. As the group presented, it was mentioned how to set up such controls without establishing a very authoritarian artists commune.

6.2.4 Group 4: The Filmmaking AI Brain Chip

Group 4 started imagining their scenario from the following question, “What if generative AI was a widely used tool within the artistic process, comparable to a pen, an instrument, or Photoshop?”. Their workshop material can be seen in Figure 18.

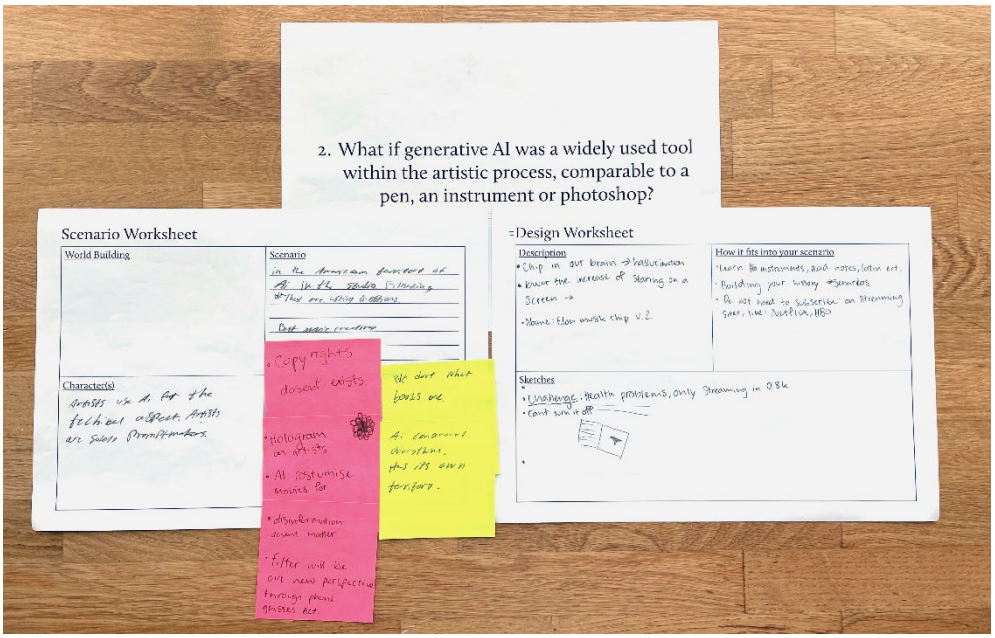


Figure 18. Material of Group 4

Group 4 imagined a scenario which is set in a film studio which uses AI in filmmaking. The AI tool has some control of the movie, customising it to get good movie reviews. The director has limited influence and the hologram projects information to the film crew as holograms. Copyright does not exist anymore.

The group designed an AI brain chip, which can create hallucinations to change how users perceive the world. It cannot be turned off once implanted, and its name is “Elon Musk Chip V.2”. In the scenario, it is used on the film

set by the director and the film crew. The resulting movie appears differently for each viewer, being automatically personalised.

6.3 Discussion

In this workshop, the participants collaboratively produced four scenarios and fictional objects that speculate about the futures of AI in art.

Although it was expected that participants would be critical towards using generative AI during the tasks, only one participant was open to experimenting with the tool. This emphasises the critical attitude of creatives towards AI, which often means a complete boycott. The one participant who used text and video generation tools expressed that he only got very biased and unimaginative responses. Furthermore, not all participants managed to document their discussions and ideas on the worksheets. However, this is to be expected and their ideas were still shared in the group presentations. Reflecting on the workshop, the participants had imaginative ideas, leading to promising openings to expand on.

Regarding grounded theory, minor additions were made to the coded data. Furthermore, through iteration and prototyping, the produced scenarios will be reworked to address the research question in a more accessible form, an exhibition.

7 Design Proposal

The following design proposal is a set of scenarios and speculative artefacts that are presented in an exhibition at the Malmö City Library during a three-week period, open to the public, which is seen as a research outcome, in line with the notion of performative research, as defined by Haseman (2006), as previously stated.

7.1 Speculative Scenarios of AI in Art

The scenarios are presented as short stories, sticking closely to the participants' material, while also making decisions to touch on relevant topics to spark further discussion. The accompanying visualisations are AI-generated using Stable Diffusion 3 and aim to provoke further critical discourse about AI in art. I recommend viewing the final posters in person or through a presentation, which can be found online[4].

[4] Link to the presentation:
https://drive.google.com/file/d/1l1IV8HWsRG1-v_f5tDft6rg-8gfOanIU/

7.1.1 The AI Muse

> *It is 2050, Tim walks into the city library, carefully listening to his AI muse through his earpiece. He is looking for a particular book which he read in his teenage years. Walking through the bookshelves, he catches the eyes of Lisa, a local artist. Hunched over her table, she quickly looks away. Tim's AI muse had told him she would be at the library, trying to get them to collaborate. Tim had always felt her distaste of his music, relying purely on human creativity, nowadays a relic from the past. But his new AI muse… it promises complexity, authenticity, and the recognition from artists like Lisa he had been craving.*
>
> *There it is, the book Tim has been looking for. His fingers trace the familiar book cover. The AI muse immediately instructs him to read specific paragraphs. As Tim reads, he is reminded of a faded memory from his late teens, just out of reach. Suddenly, chords play in his earpiece. This isn't the inspiration he imaged from an AI muse, something more… organic. It barks a command. He should return to his studio.*
>
> *He stumbles across Lisa again on his way out. Their exchange is tense, her eyes empty. He departs, not with the excitement he envisioned, but with a lingering sense of something left unsaid.*

The visualisation can be seen in Figure 19.



Figure 19. The AI Muse Visualisation

7.1.2 The Creature Generator

It is 2050, Omar steps outside his modest wooden home. Nature blooms around him. His VR headset is firmly in place. Nowadays, the majority of life takes place in the virtual world with its unlimited possibilities. Omar, a performance artist, is pioneering a new form of art using a generative AI machine. With the tool, he creates fantastical creatures and experiences their sensations first-hand. Omar transforms these experiences into performances that captivate his remote audience, enabling them to experience the bodily sensations of fictional beings. In his community, more and more people use these experiences to inspire their own art.

Today, Omar is testing a new creature design, something he hopes will be his masterpiece. He had been working

with the generative AI tool for days, as he decides to render the creature. The UI freezes briefly before the creature materializes in his virtual space. It seems tall, slender and delicate with a grotesque egg-shaped head. Eager to merge with it's consciousness, Omar proceeds to sync with the creature.

However, he experiences an unexpected and intense fear. The creature, or he himself, is in extreme panic. Omar struggles to disconnect from the virtual world. He has not experienced such anxiety. His whole body is shaking. His audience will be delighted.

The visualisation can be seen in Figure 20.



Figure 20. The Creature Generator Visualisation

7.1.3 The No-AI Commune

It is 2050, Isak waits at the border checkpoint just outside the restricted commune for artists. No AI-generated objects are allowed in, much less any AI tools. The guards scan the surroundings nervously. In the major cities, the aesthetics of basically all objects and spaces of daily life are AI-generated and most people don't care much for human creativity or authorship. Yet, Isak's curiosity about life inside the artists' commune had only grown. A world without AI interference seemed nearly mythical in contrast to AI-saturated society. Such places required strict security. There had been incidents of AI fanatics trying to infiltrate the commune to add scans of it to the "Internet of Everything", which had evolved into a spiritual quest.

As guards approach Isak, his pulse increases. One of the guards seems to be calibrating a compact, handheld device, Isak had never seen before. It has an intimidating gun-like shape. Scanning him up and down, the scanner makes a sharp warning sound, the display flashing urgently. The expression of the guards changes abruptly. Isak is confident that he has no AI-touched objects with him, entirely confused about what is happening. The guards pull an item from his bag, concealing it from his view as they discuss in low voices. Quickly, Isak is escorted away, gates closing behind him, heading down a dark hallway. His heart is pounding, filled with anxiety and confusion.

The visualisation can be seen in Figure 21.

7.1.4 The AI-enhanced Film

It is 2050, Sarah walks onto the set of the latest film she is directing. Her AI brain chip projects information about today's agenda. In her world, people depend on an AI brain chip that alters their senses in daily life. Sarah directs her actors with precision, seeing her vision of the scene clearly in front of her, a process enabled by the AI chip, which has become as common as the camera in filmmaking. As she wraps up a crucial scene, Sarah reviews the footage on set with excitement. Her vision is not just executed but enhanced by the AI through its connection to her mind.



Figure 21. The No-AI Commune Visualisation

Days later in her studio, however, she realizes the scenes appear drastically different from what she directed. Confused, she calls in her technical team, questioning the AI enhancements. The AI has been tailoring the final film to suit the audience's preferences, making it strikingly different from her original vision.

In film school, she had been taught about the potential of AI in interaction with human creativity, enabling her artistic process. Her career had flourished on this conviction. However, this latest version of the AI had been calibrated by the production company, changing its influence greatly.

Sarah questions her role as a director as the film, seemingly out of her hands, becomes a hit. It is praised

for its adaptive creativity, highly tailored to each individual viewer.

The visualisation can be seen in Figure 22.

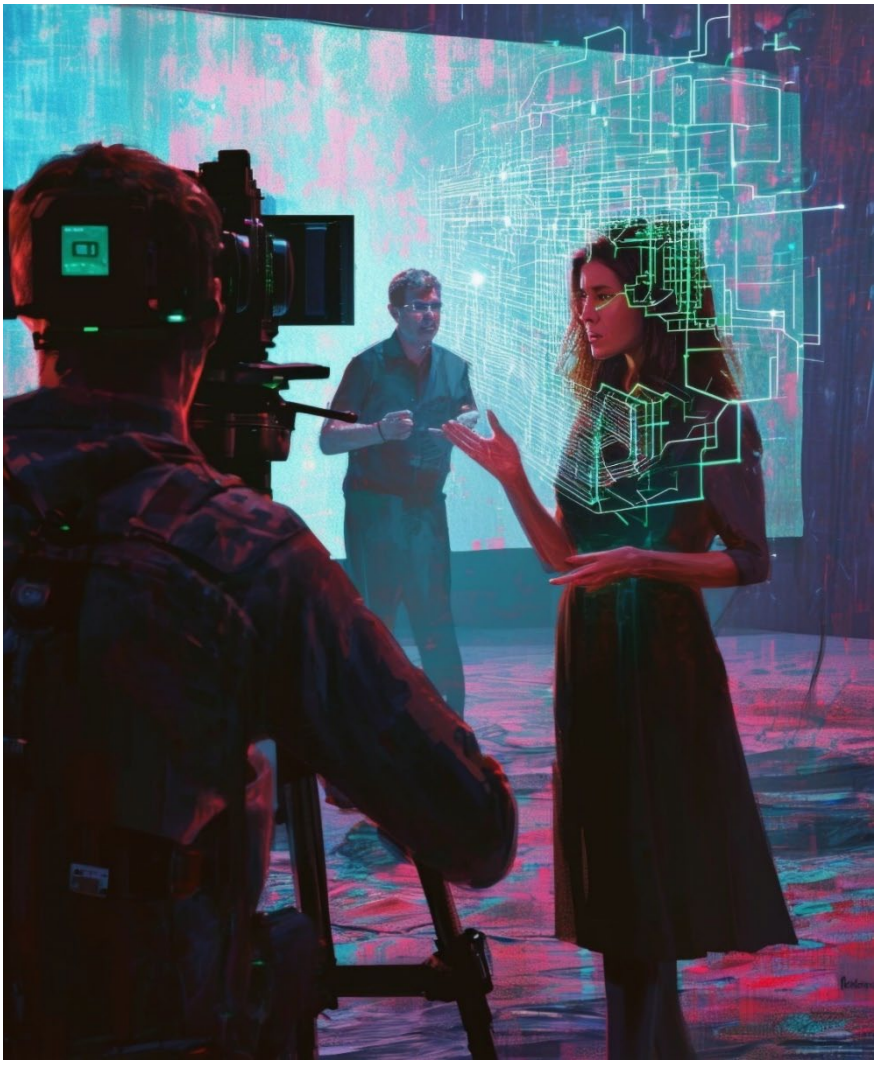


Figure 22. The AI-enhanced Film Visualisation

7.2 Related Work

Few exactly similar projects exist. However, the outcome can be compared to “United Micro Kingdoms” by Dunne and Raby (2013), insofar as it shows a set of four scenarios presented in stories and artefacts.

Furthermore, Lin and Long (2023) conducted a SD project on four themes, environment, data privacy, embodiment, and play, while generating their visualisation using generative AI, similar to the design proposal. Yet, the project does not apply a participatory approach.

Additionally, Blythe (2023) explored the use of AI-generated material in design fiction. The design proposal of this project, however, only uses AI-generated visualisations, but no AI-generated text.

7.3 Evaluation

I conducted two semi-structured interviews about reflections of creatives who engaged with the exhibition. The interviewees were participants in the speculation workshop. Interviewee 1 is a musician and Interviewee 2 draws digitally and traditionally and is an interaction designer.

7.3.1 AI-generated Visualisations

On the use of AI-generated pictures, both interviewees expressed critique. Interviewee 1 stated that these pictures were based on artists' works who were not being paid for it. Interviewee 2 said the following, "I am thinking of all the artists that had their art scraped into a database". They said that it leaves a bad feeling.

Provoking this sort of reaction was precisely the intention behind making the choice to include AI-generated visualisation. Users should be able to express their critique, get into a discussion about it, and articulate why they feel that way.

7.3.2 The AI Muse

Reflecting on the AI Muse short story, Interviewee 1 noted that the idea of a distaste for human-created art was deeply upsetting to them. Furthermore, they mentioned an AI tool which learns to inspire artists seems exciting. Interviewee 2 wondered to what extent they would still be in control of making their art, and whether or not they would feel like this art would be authentic to them. Interviewee 1 said that they "would rather have an AI tool for artists than an AI tool that compromises artists". However, it would be "creepy" that the AI Muse would have so much insight. Additionally, there were questions about who would own your data.

7.3.3 The Creature Generator

Reading the Creature Generator short story, Interviewee 1 was intrigued by the idea of art mediums of the future, asking whether or not someone living in the current time could even understand it as art, because art is such a sign of its time. Interviewee 2 asked to what extend Omar could influence the experience and what his interaction of creating the creature would look like. They stated that "AI is such a black box", while talking about whether the generator could be transparent. Furthermore, Interviewee 2 stated that sharing the techniques they use in creating their work is "half of the enjoyment in art", and that merely generating something completely loses that aspect.

7.3.4 The No-AI Commune

Both interviewees were part of the workshop group that imagined this scenario and they felt that the short story captured the idea they had in mind. Interviewee 1 stated that the "What if?" question was such an extreme starting

point, so the scenario became extreme as well. Interviewee 2 stated the following, “We took the idea of splitting off from such a society to the extreme and we ended up with fascism.”, and noted that the short story captures this well. Furthermore, both interviewees discussed whether or not the AI scanner device can actually determine whether something is AI-generated. Interviewee 2 stated that it is getting increasingly difficult to identify AI-generated art on certain occasions and that there is already a discourse online, where people need to prove that their art is hand-drawn. Interviewee 1 mentioned that they couldn’t articulate what exactly lets them identify an AI-generated picture, which they compared to a black box. Furthermore, they stated that soon it might only be AI that is able to identify AI-generated pictures, making it a strange AI versus AI situation.

7.3.5 The AI-enhanced Film

Reflecting on the AI-enhanced Film short story, Interviewee 1 noted that it can already be seen that artists are no longer in control due to the influence of capital. Interviewee 2 asked questions about who owns the AI tool, and if it could be publicly owned. Furthermore, they mentioned that such an AI tool would be great to have especially if it is part of the artistic culture, as mentioned in the short story with the film school, but only if it wouldn’t turn on artists later on.

8 Discussion

8.1 On the Use of Generative AI

The elephant in the room here is the question of whether or not one should use generative AI tools, knowing about the unethical practices common in the AI industry. Throughout the project it has become apparent that many artists chose to boycott, if not all AI tools, then often at least image generation tools. They have good reason to do so. The disregard of big tech companies for the intellectual property of artists, scraping their artworks for their training sets, has reached an extreme. Especially considering that these companies strive to exploit publicly accessible value for their private profit.

Being aware of this, the genie, that is generative AI, is out of the bottle. Large parts of the internet are scraped and this cannot be undone. In a way, a dataset involving such amounts of human creativity can be seen as a beautiful thing. However, it is terrible that this is in private hands to be exploited for profit. As Crawford (2021, p. 217) states, “The core issue is the deep entanglement of technology, capital, and power, of which AI is the latest manifestation”.

I hold that refusing to use AI tools is not beneficial, even considering the outlined circumstances, because that prevents one from investigating and critiquing AI tools intricately. There is a need for creatives to stay at the forefront of AI tools to be engaged in the discourse around it. To stay at the forefront means to interact with AI tools to experience what their interfaces afford to imagine how things can be different.

As many participants have voiced, they would like to see AI tools for artists rather than AI tools that replace artists, tools that support their creativity rather than replace it. To achieve such a future, it would be beneficial for creatives to be engaged in the use and discourse around AI tools.

8.2 The Responsibilities of Designers

Addressing designers, with the previously expressed sentiment in mind, there is a responsibility for designers engaged in the design of AI tools to strive to better address the needs of creatives. As expressed by participants, AI tools are black boxed which contradicts the need by artists to be able to manipulate their tools. AI tools ought to be transparent.

8.3 Collaborative Speculation

Reflecting on the process of conducting SD with a participatory process, it can be said that the sub-research question, “How can Speculative Design be facilitated in a Participatory Design process as to spark critical debate about AI in art among creatives?” was addressed. The workshop format of collaborative speculation produced rich scenarios that sparked discussion over a variety of themes related to AI in art.

Applying the framework of Farias et al. (2022) to assess the participants’ level of engagement in a SD project, I claim that the project exhibited a level of engagement of “collaboration” and a category of participation “shared authorship”, because non-designers produced the speculative designs, only slightly refined by the designer. The project does not qualify for the higher level of participant engagement of “leadership” and a category of participation of “initiative”, only because the initiative needs to start with non-designers, which is not the case in this project.

Furthermore, addressing the *design before design*, as defined by Pedersen (2016), the effort invested into negotiating with partners at the Malmö City Library about the location, time, and available resources for the workshops is not to be understated.

8.4 Relevance & Contribution

The project proved its relevance in three instances: (1) it has contributed to the discourse among creatives about AI and art and its role in society in the present and future, through an exhibition speculating and provoking about various themes of AI in Art, (2) it has outlined the views of creatives towards

AI in art through significant field research, interviews, and workshops, and (3) it has explored the methodological approach of conducting SD through a PD process, applying contemporary Interaction Design methods, to spark critical debate among creatives about AI. Additionally, these three themes represent the research question and its two sub-questions.

8.5 Reflection

The future of AI is elusive and difficult to work with. Radical thinking is needed. Therefore, the methodological approach of applying SD through a PD process was appropriate, as shown in the critical discourse, resulting from the project's outcome. As outlined above, the research question and the two sub-questions can be considered as sufficiently addressed by the project.

8.6 Future Work

While the format of presenting the design proposal as an exhibition was useful for communicating the scenarios to spark discussion, other forms could have different strengths that could be further explored. The scenarios could be shown in an interactive way online, rather than the current set of slides. Furthermore, focusing on one or two fictional design artefacts in the form of prototypes could engage individuals better than posters. This would open up the possibility to interact with the prototypes. The interactivity of an AI Muse headset could be faked with current AI tools, utilizing text-to-speech and language models through their API. Furthermore, an AI Scanner as present in the No-AI Commune short story could be prototyped. Interacting with its UI could lead to interesting reflections among users.

9 Conclusion

In this thesis, I explored speculative futures of AI in art, by conducting SD through a PD process, involving creatives critically throughout the project. The design process was informed by field research and interviews with creatives. Two co-design workshops were held, one exploratory and one speculative, which were hosted at the Malmö City Library. The design proposal consists of an exhibition, including four short stories and AI-generated visualisations, sparking discussion and reflection about the topic of AI in art, in line with the method of Design Fiction. The key insights of the design process and the discourse surrounding the resulting exhibition are the following. AI tools need to be investigated and critiqued through use by creatives. Designers working on AI tools have a responsibility to design for transparency in AI tools to address the needs of creatives, rather than designing a black box. SD is the appropriate way to address the wicked

problem of AI in art, especially grounded in a PD process. Furthermore, there is potential for meaningful future work in expanding on the speculative scenarios by designing physical interactive prototypes.

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